

Exercice 1

$$\begin{aligned}
 a) N &= \left\lceil \frac{\log_{10}((10^{AP/10}-1)/(10^{AS/10}-1))}{2 \log \frac{\omega_p}{\omega_s}} \right\rceil \\
 &= \left\lceil \frac{\log_{10}((10^{\frac{7}{10}}-1)/(10^{\frac{16}{10}}-1))}{2 \log(\frac{0,2\pi}{0,3\pi})} \right\rceil \\
 &= \lceil 2,78 \rceil = \boxed{3}
 \end{aligned}$$

$$\omega_p = \frac{\omega_p}{(10^{AP/10}-1)^{\frac{1}{2N}}} = \frac{0,2\pi}{(10^{\frac{7}{10}}-1)^{\frac{1}{6}}} \simeq 0,5$$

$$\omega_s = \frac{\omega_s}{(10^{AS/10}-1)^{\frac{1}{2N}}} = \frac{0,3\pi}{(10^{\frac{16}{10}}-1)^{\frac{1}{6}}} \simeq 0,5$$

$$\omega_c = \frac{\omega_p + \omega_s}{2} = \frac{0,5 + 0,5}{2} \simeq \frac{1}{2} = 0,5$$

b) N est impair donc on a $k=0,1,\dots,5$ et $\Theta = \frac{k\pi}{N}$

Il n'y a pas de zéros pour ce filtre

$$p_0 \quad \Theta = 0^0$$

$$p_1 \quad \Theta = \frac{\pi}{3}$$

$$p_2 \quad \Theta = \frac{2\pi}{3}$$

$$p_3 \quad \Theta = \frac{3\pi}{3} = \pi$$

$$p_4 \quad \Theta = \frac{4\pi}{3}$$

$$p_5 \quad \Theta = \frac{5\pi}{3}$$

†

c) $H(S') = \frac{\omega_c^3}{(S'-p_2)(S'-p_3)(S'-p_4)}$, p_2, p_3, p_4 sont les pôles stables.
Calculons la valeur de ces pôles:

$$-p_2 = \omega_c e^{j2\pi/3} = \omega_c \left(\cos \frac{2\pi}{3} + j \sin \frac{2\pi}{3} \right) = -0,25 + 0,433j$$

$$-p_3 = \omega_c e^{j\pi} = \omega_c (\cos \pi + j \sin \pi) = -0,5$$

$$-p_4 = \omega_c e^{j4\pi/3} = \omega_c \left(\cos \frac{4\pi}{3} + j \sin \frac{4\pi}{3} \right) = -0,25 - 0,433j$$

$$\text{Ainsi } H(S') = \frac{0,5^3}{(S' + 0,25 - 0,433j)(S' + 0,5)(S' + 0,25 + 0,433j)}$$

d)

$$\begin{aligned}
 e) H(s) &= \frac{(0,5)^3}{(s+0,25-j0,433)(s+0,5)(s+0,25+j0,433)} \\
 &= \frac{(b_0)/s^3}{s^3 + a_2 s^2 + a_1 s + a_0} \\
 &= \frac{b_0/s^3}{1 + \frac{a_2}{s} + \frac{a_1}{s^2} + \frac{a_0}{s^3}}
 \end{aligned}$$

